

HW section 3.1 write-up.

10. If  $m \cdot n \mid k$ , then  $m \mid k$  and  $n \mid k$

Proof: Let  $mn \mid k$ . This means  $k = mn \cdot j$  for some  $j \in \mathbb{Z}$ .

Therefore  $k = m(nj)$  and  $k = n(mj)$  where

$nj$  and  $mj$  are integers by closure of  $\mathbb{Z}$  under multiplication.  
Thus  $m \mid k$  and  $n \mid k$

□

21. Find a counterexample to show if  $m \mid a$  and  $n \mid b$  it is not necessarily true that  $(m+n) \mid (a+b)$ .

Let  $m=2$  and  $n=3$  with  $a=6$  and  $b=6$

Then  $2 \mid 6$  and  $3 \mid 6$  But  $5 \nmid 12$

Find  $\gcd(715, 517)$  using Euclidean Algorithm

$$715 = 1 \cdot 517 + 198$$

$$517 = 2 \cdot 198 + 121$$

$$198 = 1 \cdot 121 + 77$$

$$121 = 1 \cdot 77 + 44$$

$$77 = 1 \cdot 44 + 33$$

$$44 = 1 \cdot 33 + 11$$

$$33 = 3 \cdot 11 + 0$$

$$\gcd(715, 517) = 11$$

Find integers  $x$  and  $y$  such that  $1 = 183x + 103y$

By Euclidean Algorithm

$$183 = 1 \cdot 103 + 80$$

$$103 = 1 \cdot 80 + 23$$

$$80 = 3 \cdot 23 + 11$$

$$23 = 2 \cdot 11 + 1$$

$$1 = 23 - 2 \cdot 11$$

$$= 23 - 2(80 - 3 \cdot 23) = 7 \cdot 23 - 2 \cdot 80$$

$$= 7(103 - 80) - 2 \cdot 80 = 7 \cdot 103 - 9 \cdot 80$$

$$= 7 \cdot 103 - 9(183 - 103) = -9 \cdot 183 + 16 \cdot 103$$

$$x = -9$$

$$y = 16$$